bioacoustic_pipeline

A **falsifiable, multi-view pipeline for bioacoustic structure discovery.**

This is not a classifier that emits labels. It decomposes vocal audio into

acoustic s

Audio

| Corpus

Design principles

1. **No single model produces "the answer."** Each module emits an auditable

signal; s

Layout

src/bioacoustic_pipeline/

ingestion

Install (editable)

```
pip install -e ".[dev,ml]"          # core + torch/sklearn/onnx
```

pip install

Commands

Run the pipeline (audio → verdict)

python sc

Track A: engineering validation on LibriSpeech

python sc

Track B: bat (corrected, annotation-grounded)

python sc

Track B: marmoset (call-type labels)

python sc

Cross-species + representation-dependence

python sc

Determinism + reproducibility (Claim A)

python sc

Negative control (ESC-50 environmental sound)

python sc

Playback design (Phase 1: falsifiable stimuli + manifest)

python sc

Train species tokenizers

python sc

Cross-corpus DUSTED recurrent acoustic-term discovery

Marmoset (HECB sub-call units, oracle)

python scripts/hecb_dusted_marmoset.py --n-eval 1500

Bengalese finch (GT syllable-validated positive control)

python scripts/dusted_bengalese.py --birds 0 1 2

Egyptian fruit bat (bout-level, emitter vs context)

python scripts/dusted_bat.py --n-bouts 2000

Bat held-out emitter validation (70/30 split)

python scripts/bat_heldout.py --n-bouts 2000

RookID individual-ID (whole-call vs HECB, cross-session held-out)

python scripts/rookid_individual_id.py --max-calls 3000

RookID-DUSTED motif discovery

python scripts/rookid_dusted.py --max-calls 2000

Unsupervised word-structure benchmark (LibriSpeech)

§14: HECB vs HuBERT vs CPC vs fingerprint same-word consistency

python scripts/unsupervised_word_structure_benchmark.py --n-files 200

Stage C: DUSTED word-term discovery (DTW, HuBERT's native metric)

python scripts/dusted_librispeech_words.py --n-files 200

Stage D: ES-KMeans-style word-embedding clustering

python scripts/eskmeans_librispeech.py --n-files 200

§16: HuBERT sanity ablation (layer × k × collapse × pooling grid)

python scripts/hubert_ablation.py --n-files 200

Lint + test

ruff check

Validated results

Track A (LibriSpeech, 150 files)

Metric	Result
§1 combined/multi-resolution F1 (CPC	**0.836** (P=0.77, R=0.92)
§2 Semantic ceiling (token→word consiste	0.6% — tokens ≠ word identity
§3 Linear probe (word identity from CPC	35.8% / **71× lift** over 199 classes
CPC latent → speaker probe	0.56 (speaker-dominated representation)

Track B: Bat (Egyptian fruit bat, 23k segments, within-Emitter null)

Representation	context_binding nmi	Cramér's V	p
Fingerprint (deterministic physics)	0.027	0.148	0.007
Spectrogram (deterministic, baseline)	**0.038**	**0.197**	0.007
AVES (learned multi-taxa SSL)	0.030	0.162	0.007

Representation-ROBUST: 3/3 significant, within-Emitter-controlled. The

determini

Track B: Marmoset (common marmoset, 12.5k calls, 5 call types)

Representation	token↔call-type V	probe acc	p
Fingerprint (deterministic)	0.606	0.83	0.007
AVES (learned multi-taxa SSL)	**0.698**	**0.98**	0.007

Representation-ROBUST: 2/2 significant. Call types (Phee/Twitter/Trill/Tsik/Seep)

recovered

Functional aliasing (representation-dependent)

Tokenization	human alias density	bat alias density
Per-species VQ-VAE	4%	0%
Shared fingerprint	0%	17.5%

Functional aliasing **flips sign** between representations → it is **not a species

discrimina

Determinism (Claim A: implementation reproducibility)

Artifact	Run 1	Run 2	Run 3	Status
verdict.json	identical	identical	identical	
tokens.jsonl	identical	identical	identical	
motifs.json	identical	identical	identical	
(all 9 artifacts)	—	—	—	**all IDENTICAL**

Three independent subprocess runs under a fixed seed (1337), CPU, deterministic

DSP (torch

Negative control: ESC-50 (BEANS-Zero subset)

Metric	Result
Final verdict	**NON_BIOACOUSTIC_COMMUNICATION**
Pipeline outputs	NBD segments + fingerprints + tokens **c
Biological axes	functional_aliasing, motif, turn, contex
Biological-source clips (dog, crickets,	`BIOLOGICAL_SOUND_NON_COMMUNICATION_BENC
Non-biological clips (rain, engine, ...)	`NON_BIOLOGICAL_SOUND`

The pipeline detects acoustic structure (segments, fingerprints, tokens) but the

domain g

Cross-corpus DUSTED recurrent acoustic-term discovery (§11–§13, §18–§19)

corpus	regime	inventory	key finding
Bengalese finch (§12)	discrete syllable song	diverse, high-purity	fixed-cosine wmean purity **0.895**; DTW
Egyptian fruit bat (§13)	graded bout vocalisations	graded/intermediate	Emitter V **~0.46** ≫ Context V **~0.16*
InfantMarmosetsVox (§11)	discrete calls (oracle HECB)	sparse, one dominant	DTW calltype V **0.51** = 4.7× exact; de
RookID (§19)	noisy outdoor corvid calls	degenerate (one dominant)	Event V **0.22** > Source V **0.08**; mo

The difference is the corpus/repertoire, not the method — DUSTED lands where predicted

by vocaliz

Bat held-out emitter validation (§17)

metric	Train (70%)	Held-out (30%)
Emitter V	0.221	**0.375**
Context V	0.085	0.149
ratio	2.60×	**2.52×**

The emitter/caller claim **survives held-out** — not a session/file artifact.

RookID individual-ID (§18)

representation	Source V	macro-F1	held-out control
whole-call fp (44-D)	**0.384**	**0.287** (4× chance)	recording-date held-out
HECB-pooled fp (44-D)	0.321	0.263	(sub-call does NOT improve ID)

LibriSpeech unsupervised word-structure benchmark (§14–§16)

unit source	within/across ratio	NN acc	word-cluster V
fingerprint (44-D)	**0.964**	**0.147**	**0.517**
CPC-km(50)	0.954	0.191	0.436
HECB	0.976	0.089	0.392
HuBERT-km(50)	1.008 (fails)	0.052	0.338 (fails)

HuBERT is metric-dependent: fails Levenshtein/clustering but recovers under DTW

(Stage C:

Modules

Sequence-Sufficiency Gate (`diagnostics/sequence_sufficiency.py`)

Prevents

Domain Gate (`diagnostics/domain_gate.py` + `negative_control_report.py`)

Runs **be

Acoustic Fingerprint (`diagnostics/acoustic_fingerprint/`)

Physics-g

Spectrogram Bridge (`representation/spectrograms/`)

Multi-view

Segment-Representation Source (`embeddings/segment_representation.py`)

Makes the

Context Binding (`diagnostics/context_binding/`)

Annotatio

Speaker-Controlled Alias Null (`alias_detector.controlled_alias_tests`)

Functiona

DUSTED Recurrent Subsequence Motifs (`motif_miner/subsequence_motifs.py`)

Adapts sp

Playback Design — Phase 1 (`experiments/playback_design/`)

Turns diag

Determinism + Reproducibility (`determinism.py` + `hashing.py`)

Frozen-inf

Ethological Firewall (`embeddings/firewall.py` + `configs/model_metadata.json`)

JSON-drive

Runtime Layer (`runtime/`)

Device ma

Status

Track A (engineering): validated on LibriSpeech. §1 combined-F1 = 0.836.

Semanti

Session 2 — representation science + Track C freeze (§21–§29, 2026-06-23)

Unified LS word-structure (§21): same 600 words × 89 types for §14 + Stage C + D.

Finge
metric-de

New scripts: `ls_unified_rerun`, `ls_speaker_invariant`, `ls_consensus_template`,

`ls_aves_

Session 3: Architecture v1.0 — multi-paradigm validation + Track C (2026-06-23)

Architecture v1.0 freeze (12 steps, all complete)

Router au

Multi-paradigm structural validation (§33–§34)

DUSTED c

H0-CMC evidence ladder + Track C

**Evider

New modules: `diagnostics/{h0_cmc, repertoire_taxonomy, structural_alternatives,

channel_a

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